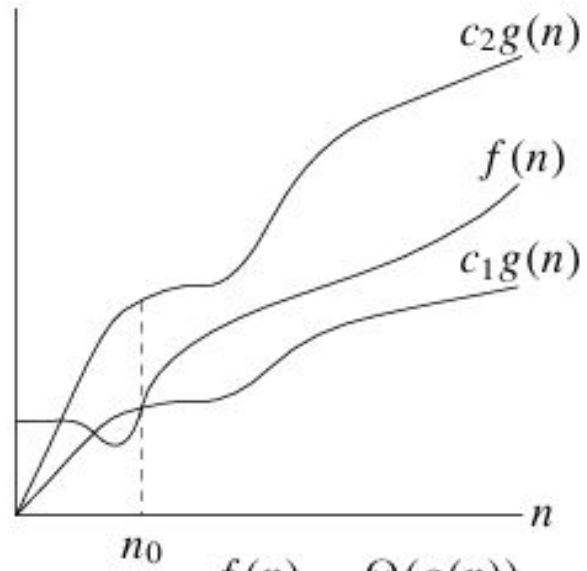


Asymptotic Notations

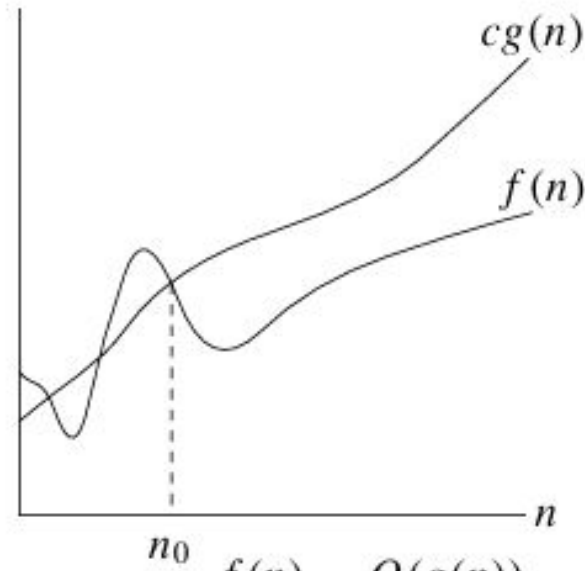
Introduction



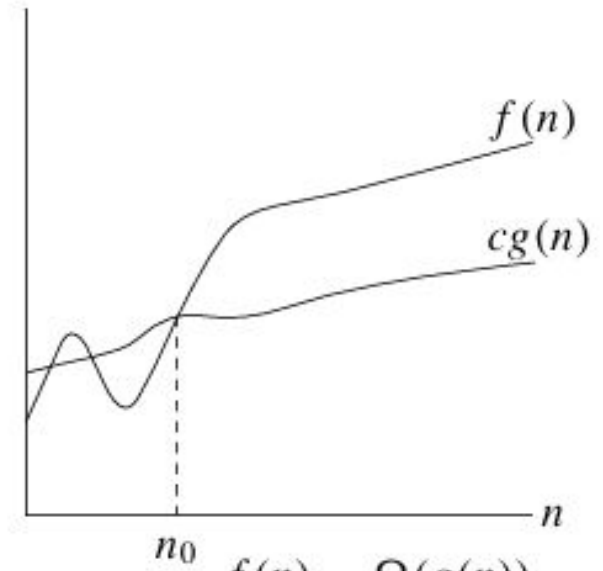
Siam Sharif Ami
Assistant Lecturer
IIUC



(a)

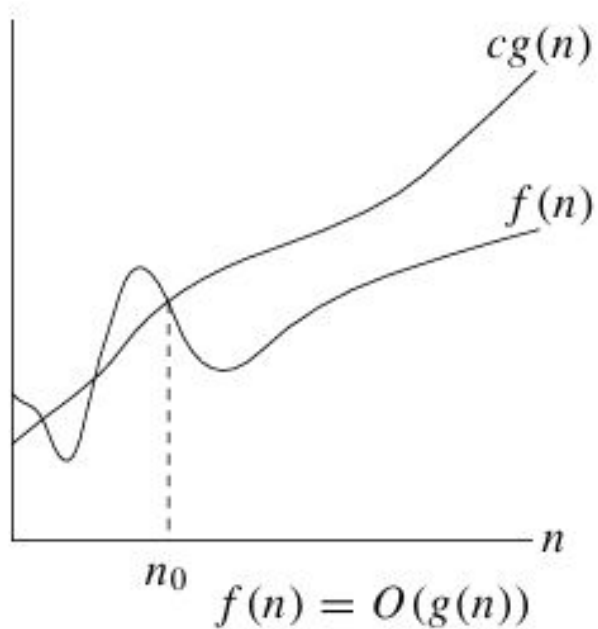


(b)



(c)

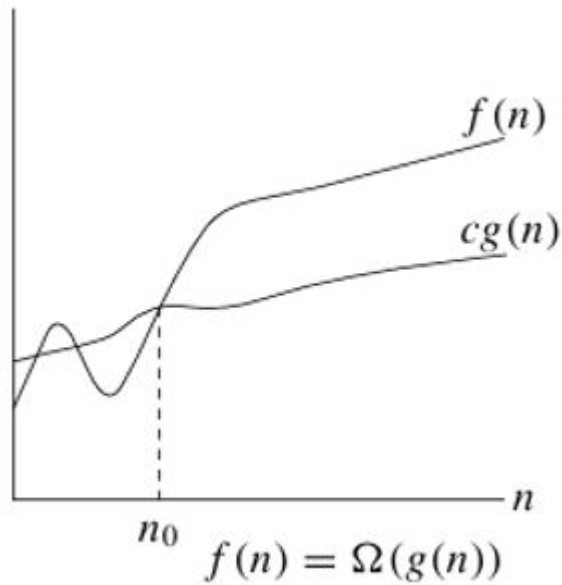
Big (O)



O- notation gives an upper bound for a function to within a constant factor. We write $f(n) = O(g(n))$

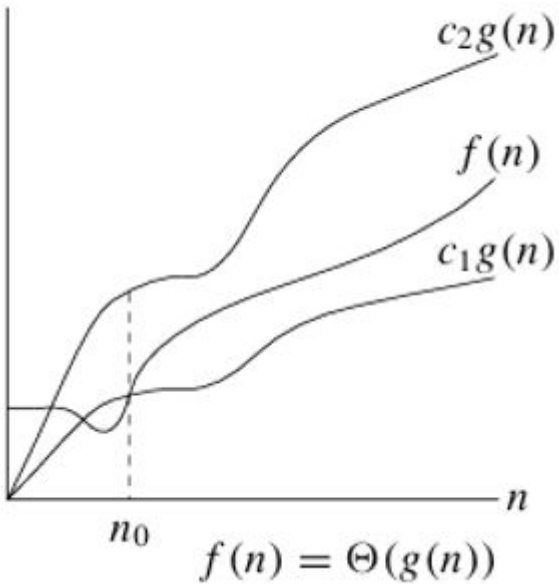
if there are positive constant n_0 and c such that at and to the right of n_0 , the value of $f(n)$ always lies on or below $cg(n)$.

Ω (Omega)



Ω - notation gives a lower bound for a function to within a constant factor. We write $f(n) = \Omega(g(n))$ if there are positive constants n_0 and c such that at and to the right of n_0 , the value of $f(n)$ always lies on or above $cg(n)$.

$\Theta(\text{theta})$



Θ - notation bounds a function to within constant factors. We write $f(n) = \Theta(g(n))$ if there exist positive constants n_0 , c_1 , c_2 such that at and to the right of n_0 , the value of $f(n)$ always lies between $c_1g(n)$ and $c_2g(n)$ inclusive.



Types of Asymptotic Notation

Asymptotic runtimes:

How does runtime scale with input size

$O(\text{Big } O)$ - **Worst Case, Upper Bound**

$\Omega(\text{Omega})$ - **Best Case, Lower Bound**

$\Theta(\text{theta})$ - **Average Case, Tight Bounds**



Approximate Runtimes

	n	$n \log n$	n^2	2^n
$n = 20$	1 sec	1 sec	1 sec	1 sec
$n = 50$	1 sec	1 sec	1 sec	13 day
$n = 10^2$	1 sec	1 sec	1 sec	$4 \cdot 10^{13}$ year
$n = 10^6$	1 sec	1 sec	17 min	
$n = 10^9$	1 sec	30 sec	30 year	
max n	10^9	$10^{7.5}$	$10^{4.5}$	30

Time complexity chart



Big-O Complexity Chart

